

Technical Memo: Costing Requirements and Quotation Guidelines for 'Varsity Cyber Clash'

TO: Mustafa (Costing & Procurement Lead) FROM: Senior Network Architect & Infrastructure Procurement Consultant
DATE: October 24, 2025 SUBJECT: Hardware Specifications and Market Quotation Requirements for Tournament Appendices

1. Memo Purpose and Strategic Significance

I am tasking you with the collection of real-world market quotations for the upcoming "Varsity Cyber Clash" E-Sports tournament. As we prepare for the 2025/2026 session, our procurement strategy must balance technical excellence with fiscal responsibility.

It is critical to note that the Appendices section, which will serve as the repository for these quotations, constitutes 30% of the total project grade. Our objective is to architect a high-performance network capable of supporting over 400 concurrent users—encompassing competitive gamers, event administrators/broadcasters, and spectators—while strictly adhering to the "Minimal Cost" mandate issued by the University Student Council.

2. Category 1: End-User Gaming Stations & Display Systems

Our procurement strategy for end-user hardware must prioritize competitive integrity. To satisfy the minimal cost objective, I suggest sourcing previous-generation components that still meet E-Sports benchmarks rather than the absolute latest releases.

Instruction: Source quotations for the following, prioritizing desktop towers over laptops for superior thermal performance at a lower price point.

- **Gaming Workstations:**
 - **Processor: Multi-core CPU (e.g., Intel Core i7 or equivalent) capable of high clock speeds.**
 - **RAM: Minimum 16GB.**
 - **GPU: Dedicated graphics (NVIDIA RTX 30-series or equivalent) to maintain frame stability.**
 - **Storage: High-speed NVMe SSD for rapid game and system loading.**
- **Display Systems:**
 - **Monitor: E-Sports standard high-refresh-rate panels (Minimum 144Hz) with 1ms response time to ensure latency does not impact tournament outcomes.**

3. Category 2: Core and Distribution Networking Hardware

Following the hierarchy required for a wired LAN in an enterprise environment, we will standardize our infrastructure on D-Link hardware as specified in our technical standards (LR7).

- **Core Switch: L3 Gigabit Managed Switch**
 - **Technical Justification: This will serve at the network core as the central network router and aggregation device. It is an ideal solution for large IP routed networks where multiple dynamic routing devices are**

present, specifically providing support for dynamic routing protocols such as RIP and OSPF, alongside advanced QoS and Security.

- **Distribution/Edge Switches: L2 Gigabit Managed or Smart Switches**
 - **Technical Justification: These units will act as aggregation points for multiple fast Ethernet edge switches and end-user clusters. They must support VLANs and provide added network security to segment the various user groups (gamers vs. spectators).**
- **Routers:**
 - **Technical Justification: Dedicated hardware is essential to manage our private IP addressing scheme and provide stable external backhaul for tournament live-streaming.**

4. Category 3: Wireless Infrastructure (Wi-Fi 6)

To support the high-density spectator zones and mobile administrative staff, we must deploy Wi-Fi 6 (802.11ax) technology.

Instruction: Sourcing at least two different quotations for Wi-Fi 6 Access Points (APs) is mandatory. The selected APs must be rated for high-density environments to prevent signal degradation in the main stadium/hall area.

5. Category 4: Server Farm Requirements

A dedicated server area is required to host the essential services defined in the project specification. Please ensure you provide hardware quotes for Rack-mount Servers capable of hosting:

- **[] Game Servers: Local low-latency hosting for tournament play.**
- **[] Web Server & Registration System: For participant check-ins and public brackets.**
- **[] File Transfer Protocol (FTP) Server: For distribution of game patches and media assets.**
- **[] Live Streaming & Tournament Management Systems: Dedicated platforms for broadcasting and bracket management.**

6. Category 5: Specialized Software, Support, and Logistics

Our costing must be comprehensive to avoid mid-event budget shortfalls.

- **Software Licensing: E-Sports management licenses and game-specific commercial licenses.**
- **Installation Materials: Source prices for bulk Cat6 cabling. To minimize costs, our team will be performing custom cable construction in-house (as per our Support Portfolio requirements).**
- **Cooling and Infrastructure: Per Source 2.2, we must account for the thermal load of 400 users and 200+ PCs. Find quotations for Portable Industrial Cooling/HVAC Rental.**

- **Operational Costs:** Internet subscription (event duration), electrical contractor setup, and estimated manpower.

7. Evidence and Documentation Standards (Appendices Checklist)

To secure the maximum score for the Appendices, every item you quote must meet these three mandatory criteria:

- 1. Real-World Price Evidence:** Every price must include a screenshot from a local vendor or a direct, functioning URL to a retail listing.
- 2. Date Validity:** All pricing data must be current and relevant to the Session 2025/2026 context.
- 3. Minimal Cost Justification:** Provide a brief note for each item explaining why it represents the best value for money without sacrificing the performance required for a regional E-Sports event.

8. Closing Directive

The complexity of this project and our group's professional reputation depend on the accuracy of your procurement data. I expect a draft of all gathered quotations for review no later than Friday, 5 June 2026. This ensures we have a one-week buffer to finalize the report before the ULearn submission deadline on Sunday, 14 June 2026 (12:00 PM).

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